

The Vibe Theory Model

The discrete universe program as it stands

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Abstract

The program models the universe as one four-dimensional hyperbolic honeycomb, $\{3, 4, 3, 4\}$, whose every site holds 24 ternary values updated by one deterministic, reversible, integer rule, with physics read on a flat three-dimensional surface at its cusp. This paper states the model as it now stands, in one pass: the base, the rule piece by piece with the reason for each, what the rule produces against computed controls, and what is still open. Since *The Missing Amplitude* the rule has gained exact amplitudes in $\mathbb{Z}[\omega][1/2]$, a working vacuum, a compact $U(1)$ light, and a gravity found by summing one trit per link, which bends light by 2, travels at c , forms horizons at $2GM$, obeys an area law, and whose spin 2 is forced by the unlabeled docks. The central new result is a no-go and a way out: on the rule every vibe is a lineon, line-only motion contradicts observation by some 17 orders of magnitude, and a stated rule change (a filled sea, a fermionic dock mixer, a swap coin, and the prime 7) gives every excitation one limiting speed and exact relativistic motion in 3d. Binding under that change reduces, through a chain of theorems, to one dynamical question about a 120-element gauge register on the husk. A changelog closes the paper. The suite holds 1,474 experiments, every number here cites one, and every failure is kept. Claude Code was used throughout, with Claude Opus 5 and Opus 5.5, to write the code and this paper.

Scope

This page is the paper in brief. Each line names what it rests on, and the sections after give the numbers.

Filled in since *The Missing Amplitude*

- **The missing amplitude is supplied.** The rule now carries exact amplitudes in $\mathbb{Z}[\omega][1/2]$. The meeting (the fear beat) is the one non-Clifford gate, and a moving vibe is quantum while a static one is classical (Section 3).
- **A rule with a reason for every piece:** the $W(F_4)$ knit, the collision, the doublet lock, the meeting, the pass, the no-veto store and the coin, each chosen over a stated alternative (Section 2).
- **A working vacuum**, balanced, reversible and covariant on 51 of 51 cases (Section 2).
- **Light**, as compact $U(1)$ from column sums, with two polarizations and Coulomb's $1/(24\pi r)$ (Section 3).

- **Quantum results on the current rule:** Bell, Tsirelson, contextuality as fear, no-cloning, teleportation, GHZ, spin-statistics, Pauli exclusion, and the second law with its arrow (Section 3).

Solved

- **Gravity as a mechanism.** One trit per link, summed into a depth, gives $1/r$, the same fall for everything larger than a dock, light bending by 2, a pull at c , and α held constant (Section 4).
- **Horizons:** a radius of $2GM$, Hawking’s redshift scaling, particles from a forming horizon, and an area law with Bekenstein’s bound as dynamics (Section 4).
- **Spin 2 derived, not chosen:** sliding the unlabeled docks forces linearized Einstein–Hilbert, the ADM structure, a wave speed of exactly c and $\lambda = 1$ (Section 4).
- **Randall–Sundrum’s short-range number** in the flat bulk, with its $4/3$ factor derived (Section 4).
- **A composite at relativistic speed** on a line, and falling alike (Section 8).
- **Motion in 3d, one-body.** Line-only motion is ruled out by observation. A stated rule change gives every excitation one limiting speed $c^* = c/2$ and exact relativistic, isotropic motion, in the many-body rule, with the vacuum inert and gravity’s sign kept (Section 5).
- **The binding problem, narrowed by theorems:** the band sum, the Kesten floor, theorems A and B, the first order walls of $\mathbb{Z}_3$ and of the bulk, and a $2I$ string that loosens continuously on the husk (Section 6).

Still open

- **Binding that moves.** Whether the beat keeps the ordered $2I$ vacuum long enough, and then a light bound member moving in 3d with inertia equal to energy. This is the program’s sharpest open question.
- **A Planck temperature** for horizons, and the bulk’s warp, radion and brane bending.
- **One light speed** for the husk light on the candidate rule.
- **The electron across lines and g , the weak force and CP violation, the value of α , a bounded self, and one rule for every row** (Section 7).

The candidate rule of Section 5 is not yet the working rule: it is adopted only once binding works on it.

1. The Base

The program’s aim is to find a discrete base from which space, time’s arrow, relativity, spin, light, the quantum, gravity and selves each come out, or to show the base is wrong [7, 9]. It draws on Margenstern’s cellular automata in hyperbolic spaces [11]. Every claim below is one experiment in the suite, cited by its code (E-SPN-0143 and so on), each with gates fixed before its run and a computed control. A result is graded by depth: L1 is algebra, L2 a model run on the mesh, L3 the rule itself.

1.1. Terms

- **Tone.** One trit, $+1$, 0 or -1 . The whole state is tones. Three is the smallest alphabet with an empty value and a sign flip, which is charge conjugation (E-FND-0038).
- **Vibe.** A slot holding $+1$ (a *love*) or -1 (a *fear*). A slot holding 0 is calm. Charge is love minus fear, content is love plus fear.
- **Dock.** One site: 24 slots, one per root of D_4 , the 24 vertices of a 24-cell. Opposite slots pair into 12 *lines*, and the lines fall into three *frames* of four orthogonal lines. D_4 is the one 4d root system with triality, and its 24 directions carry the spinor double cover, so a 2π turn gives -1 (E-SPN-0042).
- **Mesh.** The space of docks. In the model it is $\{3, 4, 3, 4\}$, whose vertex figure is the flat cubic honeycomb $\{4, 3, 4\}$. It unfolds from one 24-cell in shells of 1, 24, 456, 8,376 and 153,192 docks, a ratio settling at 18.278 (E-GMT-0027). Experiments run on a flat periodic D_4 box of side 8, 16 or 24, a disclosed stand-in with the same 24 directions per dock.
- **Beat.** One tick. Every dock applies the rule at once, so a run is reproducible bit for bit.
- **Take.** Each beat a slot takes its neighbor's value along its root, the *stream*. Nothing moves: a value is read and written, never carried, so the stream is a permutation and runs backward exactly.
- **Husk.** The three-dimensional surface where physics is read, the depth-even half of the box, exactly closed (E-FRC-0168). In the honeycomb it is the flat horosphere at a cusp, and a spectral dimension near 3 is read there (E-GMT-0025).
- **Bulk, depth.** The 4d mesh behind the husk. A husk column is the stack of bulk docks behind one husk dock, and its size D is its depth. Every husk field is a column sum of bulk trits (E-FRC-0207).
- **Role, link.** Each vibe carries a point on a 3 by 3 grid, a qutrit. A link between two docks holds one of the 648 elements of the qutrit Clifford group $\Sigma(648)$ (E-QTM-0117). A link stores no matter: it is the relation between the vibes on it.
- **Store.** A dock line's held love and fear pair, made from an empty line and released back to it. This is how the vacuum makes and unmakes pairs with its charge fixed at 0.
- **Amplitude.** The state is a finite sum of configurations, each weighted by an exact number in $\mathbb{Z}[\omega][1/2]$, $\omega = e^{2\pi i/3}$. Every three in the model lives in this ring, and nothing is rounded.
- **Path.** One measurement record of the quantum rule: at every split one branch, chosen by a deterministic key. Fields are read as averages over paths.

1.2. The engine and the screen

The claim has the shape of a video game. A small fixed engine updates a large state tick by tick, and what anyone sees is a lossy projection of that state. Players learn a game's physics from the screen without reading its code, and physics is that study of the screen. The program is a guess at the engine, one exact rule with no dice, tested by running it and checking that it draws the screen we measure. The engine is the rule, which

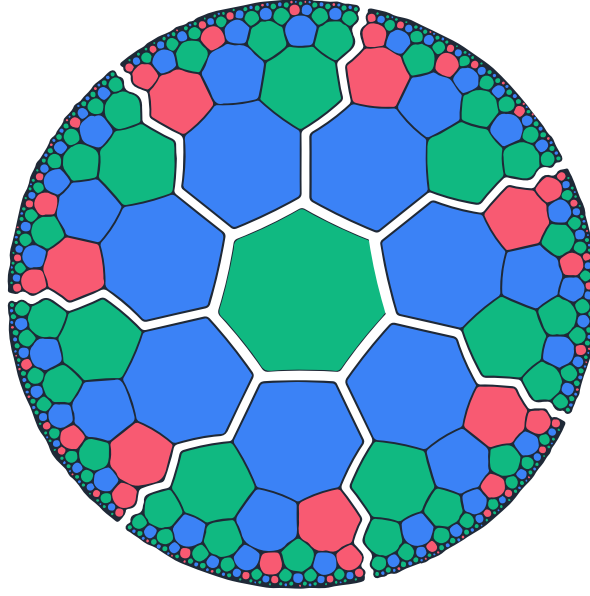


Figure 1: The $\{7, 3\}$ tessellation of the hyperbolic plane on the Poincaré disk, a two-dimensional stand-in for the four-dimensional $\{3, 4, 3, 4\}$, which cannot be pictured. Room opens faster than in flat space, which is what lets a region's boundary act as a screen. The three colors mark the three values of a tone.

runs on every dock. The bulk is part of the state, and the screen is the coarse-grained husk (Figure 3). A coarse picture is not a false one. The picture stops at the hardware: the mesh runs on nothing, and nobody plays.

1.3. Grading

The working ledger lists 191 facts the model must produce, from the Born rule to the specious present, each graded under one set of rules: read on the husk, the rule integer and reversible, a stand-in labeled and never graded as a derivation, a numerical match counted only through a pre-registered null (E-MTH-0010), and the verdict judged over 17 link starts (E-MTH-0028). Today 36 are held, 52 partial, 45 stand-in, 26 open and 32 not started. No random number enters any run: starts, controls and nulls use Weyl sequences or enumeration.

2. The Rule

Each beat applies the coin, then the meetings, then the collision, then the stream. Every piece is an exact, reversible map in $\mathbb{Z}[\omega][1/2]$ with no float, no rounding and no random number, and runs backward piece by piece. Each piece below is the one that survived a stated alternative.

- **The knit** is the whole set of pieces, and it keeps $W(F_4)$, the full symmetry of the dock (1,152 elements). Such a knit exists and keeps CPT, so no direction of the dock is preferred (E-RLT-0061).
- **The collision.** A dock's full lines turn as -1 and a lone vibe goes straight through, so a lone vibe's wake stays on its own line, the free particle's straight path (E-RLT-0084).

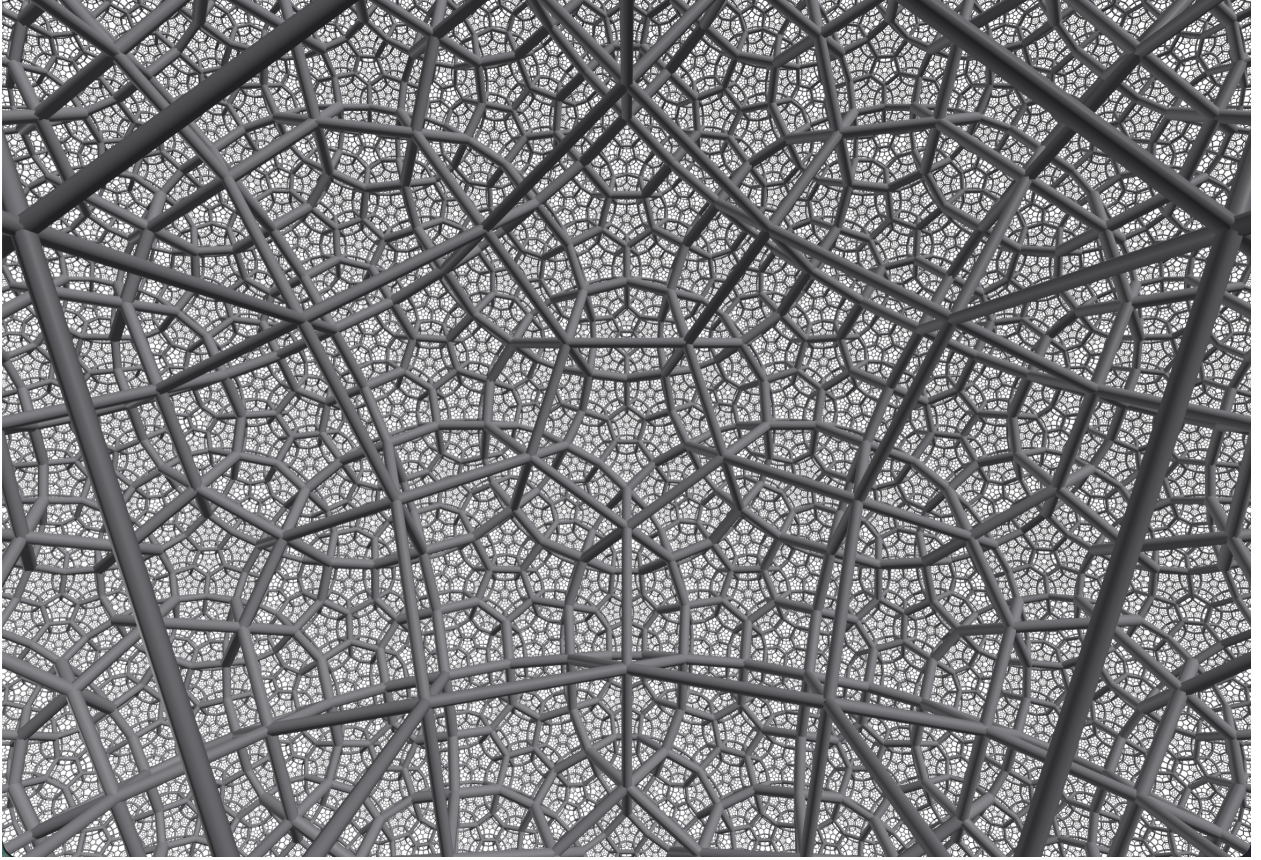


Figure 2: Inside the $\{5, 3, 4\}$ honeycomb of dodecahedra, the closest cousin that can be rendered in three dimensions. It is a control, not the subject: its twelve directions carry no spinor, where the 24 directions of the committed $\{3, 4, 3, 4\}$ do.

- **The doublet lock.** A vibe's take direction along its line is its role's spin doublet. It is the only way a vibe can move with amplitudes and stay covariant, and it makes every charge-one state a spinor (E-RLT-0097, E-SPN-0071).
- **The meeting.** A love and a fear on one line pass unchanged. Two like vibes keep their points (weight $1/4$) or exchange them ($3/4$), leaving a *knot* with CHSH $\sqrt{7}$ (E-RLT-0099).
- **The contact.** Two like vibes on a full line keep their slots, the *pass*, phase $+1$, instead of bouncing, -1 . Of the six phases the ring allows, only these two add no new number, and only the pass binds the electron (E-SPN-0092, E-SPN-0093).
- **The store.** Pairs are made and released with no veto, each stored pair holding two points. It is the only vacuum measured that holds under the pass (E-RLT-0102, E-RLT-0103).
- **The coin.** On a line holding one open vibe, $2C = (1 + \omega)I + (1 - \omega)X$ on its two slots. It gives a lone vibe a mass, with a gap of $2\pi/3$ and a top speed of $1/2$ a beat, and its cost is exact lattice momentum, the one symmetry that forbids a mass (E-SPN-0090, E-SPN-0091).

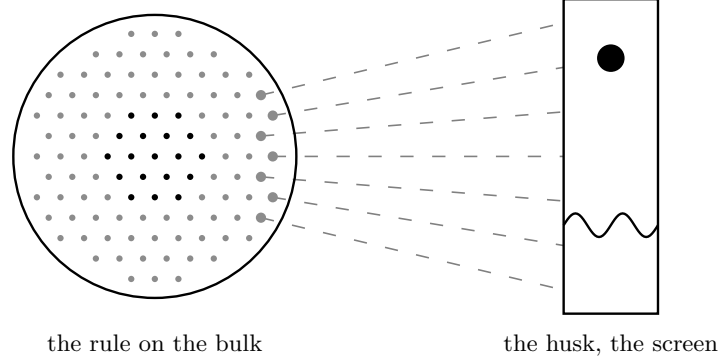


Figure 3: The engine and the screen. The rule runs on every dock of the bulk, drawn as a ball, and its activity reaches the husk along rays. On the husk, the screen, it shows as what physics measures: a particle and a wave. On the site the docks switch only on a beat while the projection flows. This is a still.

2.1. The working vacuum

The vacuum is the no-veto store under the pass, with the coin on, and one C domain with no walls. On 51 of 51 gated cases (E-RLT-0105) it keeps its conservation laws, because no vacuum vibe is ever alone on its line and the coin acts on it 0 times on every path. It runs backward exactly, keeps C, and makes every husk dock one causal world. Pairs made equal pairs unmade, 466,944 each, on a cycle of 12 beats (E-GRV-0072). A lone love spreads over 105 to 124 occupations by beat 16 on 17 of 17 starts, so positions carry amplitude at no cost to the vacuum. Not held: the kept branch of a like meeting does not interfere, and the lone love reads the dephased $1/4$, $1/16$, $1/64$.

2.2. The line law

With no mixer, the tone on every straight mesh line is conserved: 6,144 of 6,144 lines on side 8 over 48 beats (E-SPN-0098). Every piece acts inside one line, so the knit is 6,144 one-dimensional worlds that never touch. Line locality is momentum conservation: a lone vibe's momentum is its root, so it cannot change line without changing momentum. The vacuum is therefore correlated without limit along each line and not at all across lines, and three failures are one fact: no $1/r$ response (E-GRV-0086), no area law (E-GRV-0083), and no quiet reach across lines (E-SPN-0101). Every mixer the rule's ring allows cascades through this vacuum (E-SPN-0094 to E-SPN-0099), and no store can join lines, since an involution must return every vibe to its slot (E-RLT-0107). Section 5 is where this is resolved.

3. Light and the Quantum

3.1. Light

Light runs beside the knit on the links. Every husk angle, flux and counter is a column sum of bulk trits, with coupling $\kappa = 2/(2D + 1)$ (E-FRC-0207). A bounded, reversible column of $2D + 1$ values is a circle, so the light is compact $U(1)$ exactly (E-FRC-0244, E-FRC-0245). On the husk it has exactly two polarizations per momentum, the bulk's third killed by the projection (E-FRC-0179). A charge's potential is $1/(24\pi r)$ with no $1/r^3$ term, its quartic isotropic by an integer identity (E-FRC-0241). Charge is counted in whole units locally on every beat, pair creation included (E-FRC-0243). Matter reads the light as an exact Peierls phase, with

like charges pushed apart at 0.96 to 1.17 of the static Coulomb prediction, on one line with no back-action (E-FRC-0252).

3.2. Where the quantum comes from

Without the meeting the knot is Clifford and can be simulated classically. The meeting, the *fear beat*, is the one gate outside the Clifford group (E-QTM-0118). Under the lock, motion adds a second source: no locked direction is a stabilizer state, so a moving vibe carries a fixed fear share of 0.164 (E-RLT-0099). A static vibe is classical and a moving one is quantum.

On this footing the following hold on the current rule, each over 17 starts where a start applies:

- Bell violation: CHSH above 2 on exact algebraic values, never 2 (E-QTM-0140).
- Tsirelson's bound: the exact largest CHSH of a knot in closed form, and $2\sqrt{2}$ reached by a finite history (E-QTM-0132, E-QTM-0133).
- Contextuality is the fear: a knot is contextual for the model's own readings exactly when it holds a fear, with total deficit $18(e^{\text{mana}} - 1)$ (E-QTM-0149, E-QTM-0150).
- No-cloning, teleportation, swapping and GHZ: no beat duplicates an unknown signed state, and a love and fear pair met once teleports and swaps exactly (E-QTM-0151 to E-QTM-0153).
- Spin-statistics: a full turn is $(-1)^N$ with N the doublet count, so turn sign equals exchange sign (E-SPN-0071). Pauli exclusion holds as one vibe per slot, in the base and in the dynamics (E-SPN-0045, E-SPN-0047).
- The second law and the arrow: coarse entropy rises a thousandfold forward and under the exact inverse alike, while fine-grained entropy stays exactly constant (E-FND-0146, E-FND-0147).

Measurement is bounded by a theorem: a frame-covariant rule depolarizes every basis at one rate, since its roots form a 2-design, so a pointer basis must come from the environment and never from the rule (E-QTM-0154).

4. Gravity

A gravity has to fall as $1/r$ (a massless field), only attract (even spin), act the same on everything (a charge-blind source), arrive at c (GW170817 put it within 1.7 s over 130 million light years), bend light by 2 (clock and space stretched equally), and carry positive energy. Everything read off the rule and its light alone failed: entropic pull, time dilation near a held knot, occupancy-gated takes, a neutral lump in the depth, growth, shared history (E-GRV-0056 to E-GRV-0073), the light's second-order residue, which is van der Waals at $r^{-6.5}$ (E-FRC-0253), and a second light sourced by content, which repels like sources as spin 1 must (E-GRV-0074).

4.1. The depth

What works is the husk's depth. A deeper column holds more docks, so the same count slows a clock and lengthens a span: the arena changes, not a force on a charge, and so it is the same on everything. As a spin-0 wave sourced by content, the *radion*, it attracts as $1/r$ with no chosen sign and is causal (E-GRV-0079, E-GRV-0080). The depth is not stored. Each link holds one trit, a step up, level or down, and steps out minus steps in equals content at every dock, so a lump's lines thin as $1/r^2$ and the summed depth falls as $1/r$ by geometry: Gauss exact on 43,008 checks, the sum equal to the radion to 5.6×10^{-17} (E-GRV-0090).

A link divides by the depth count of the columns it joins, $q = 2D + 1$, so light's index is q , a clock slows as $q^{-1/2}$ and space stretches as $q^{1/2}$. Light bends by a factor 1.999 and 1.995 read from the exponents, and 2.258 through a lens against general relativity's 2.338 and the clock-only control's 1.091 (E-GRV-0092, E-GRV-0093). Holding the gauge field's resolution at one depth everywhere, as minimal coupling does, keeps α flat to 2.8×10^{-8} (E-FRC-0257, E-GRV-0099). A calibrated rod front times the pull at $1.0023c$ on the husk and never above light on the warped stack (E-GRV-0104). The step field is gravity's own field, as the Coulomb field is light's, sourced by the rule's energy: the confining string is the energy-line register mod 3 with Gauss exact (E-GRV-0127), and no local move on a closed register turns those lines into the $1/r$ depth (E-GRV-0130, E-GRV-0133). On a love sea, normal-ordered energy fixes the sign, a hole sourcing $+1$ times a love's depth, with the conserved register per slot the count minus twice the charge, both from the sea (E-GRV-0144).

4.2. Horizons

Where the clock register runs out of room, time stops and light is trapped, exactly. The radius is $2GM$ and grows as M once the box is taken out, slope 0.994 to 1.012, with bending 1.9985 and α flat (E-GRV-0118, E-GRV-0122). Light leaving a large horizon is redshifted at 0.97 to 0.98 of its surface gravity κ , and TM is flat to 1.7% (E-GRV-0123). A forming horizon makes particles, exactly counted (E-GRV-0132). Carrying the infall onto the horizon's own surface registers hides 18.18 nats per cut link at every size and every stage of growth, and a horizon forms only while what fell in fits on its surface: Bekenstein's bound as a rule of the dynamics, with $S/(A/4G)$ a flat 1.44 to 1.47 (E-GRV-0131, E-GRV-0135). The 3.8% energy drift at a depth boundary was the wrong energy being read: the headroom light keeps the right one to 2.9×10^{-14} and bends by 2.0069 (E-GRV-0126, E-GRV-0128).

4.3. Spin 2 from unlabeled docks

One depth per line class fixes a full metric carrying exactly two polarizations (E-GRV-0125). The coupling is not chosen. The docks carry no labels, so sliding where they sit changes nothing, and that symmetry alone forces linearized Einstein-Hilbert (E-GRV-0138). Slid in space and time it makes the lapse and shift Lagrange multipliers with DeWitt's kinetic term (E-GRV-0139). At cubic order the full slide fixes the wave speed at exactly c and $\lambda = 1$, which a slide keeping the time slicing leaves free (E-GRV-0141).

4.4. The bulk

A shrinking bulk, $\{3, 4, 3, 4\}$ seen from its husk, keeps $1/r$ at 0.549 of the husk alone's strength with a positive short-range correction (E-GRV-0100), as in Randall and Sundrum [10]. Finer layering takes the scalar to $3/4$ of their short-range number (E-GRV-0137). With the slide-derived spin 2 in the bulk the tower is the scalar's

times a Gram matrix, reaching 0.971 of it and 0.998 from the kernel (E-GRV-0142), and a slide along the depth forces 5d Einstein–Hilbert, which derives the 4/3 exactly (E-GRV-0143).

The grade is L2: the step field is a part added to the rule, and the factor 2 follows from the span change by construction, which the runs show the integer rule realizes.

5. Motion in 3d

5.1. Every vibe is a lineon

The line law makes every vibe mobile only along its line, and composites inherit it exactly. Composites on parallel lines are independent (E-SPN-0117). A pair bound across crossing lines is held at the crossing whatever its string does (E-SPN-0116).

Theorem 1 (The star) A composite whose lines all pass through one dock is pinned there at every order, because the only line-changing piece acts at that dock alone (E-SPN-0118, E-SPN-0119).

Two such hubs set off a cascade through the vacuum, and binding them with a string does not stop it, since a string that is a phase moves nothing (E-SPN-0120). No fused mover exists (E-SPN-0125), and the bulk’s shadow adds no motion (E-SPN-0127). The six other routes tried, from borrowing background matter to a vacuum of mobile line pairs, each fail with a stated cause (E-SPN-0121 to E-SPN-0137).

5.2. Line-only motion is excluded by observation

Averaged over line classes, a lineon’s band is isotropic through P^4 by the 5-design, but its first direction-dependent term is a series in P/Mc , not in Pa , so the anisotropy stays at every dock size. Its inertia is 4.02 times its rest energy, so its effective 3d speed is $c/2$, and a massless excitation’s speed depends on direction by 20%: $0.354c$ along an axis, $0.417c$ along a face diagonal, with velocity up to 36° off momentum. The laboratory isotropy of c holds to about 10^{-18} , so line-only motion is excluded by some 17 orders of magnitude, and a beam would spread sideways as fast as it moves forward (E-SPN-0126). The rule must change.

5.3. A candidate change

The change has four parts, each forced by a failure of the one before.

- **A filled love sea as the vacuum.** Any vacuum of line pairs that matter can unpair keeps the cascade’s source (E-SPN-0137). A sea full at every dock has nothing to unpair, and a hole in it is the particle.
- **A fermionic mixer over all 24 slots of a dock.** On a full dock it is exactly blocked by Pauli exclusion, so the sea stays one branch for 128 beats and the collision never fires (E-SPN-0140). A mixer over one frame’s 8 slots confines velocity to that frame, reaching $0.5c$ elsewhere (E-SPN-0136). The 24-slot mixer’s hop is $(e^{i\theta} - 1)/24$, so it adds one number, $1/3$.
- **The swap coin** ($\zeta = -1$: keep 0, cross 1, integer entries). Every other coin lets a lineon or the massless pair outrun the rest: a lineon tops out at $\cos(\varphi/2)c$ and the massless pair at $c/2 + 2\gamma K$, $\gamma = \cot(\varphi/2)/8$ (E-SPN-0143).

- **The prime 7.** A norm-one unit of $\mathbb{Z}[\omega][1/6]$ is a sixth root of unity (Hilbert 90), which pins the mixer's angle and leaves the lightest particle at $R = 1.1027$. Inverting $7 = N(3 + \omega)$ admits $u = (8 + 5\omega)/7$, whose angle is an irrational multiple of π , so the mass is tunable. The ring becomes $\mathbb{Z}[\omega][1/42]$.

5.4. What it gives

Theorem 2 (The 24-cell bound) Under any dock-local mixer a branch's velocity is the average of the hops it takes (Hellmann–Feynman), so it lies inside the 24-cell of the roots. The best isotropic speed is $c/\sqrt{2}$, and a singlet spread over all 24 roots moves at exactly $c/2$, the rms of $u \cdot r$ over a 4d 2-design (E-SPN-0140, E-SPN-0142).

The hole's uniform (singlet) level sits alone at the band edge for every angle past the working one. Its inverse mass is isotropic to 3.7×10^{-10} with its first warp at K^6 , as the 5-design forces, and its dispersion is relativistic with $c_{\text{eff}}^2 = m \cot(m)/2$ and inertia over rest energy $R = \tan(m)/m$, 1.0007 at $m = 0.047$ (E-SPN-0142, E-SPN-0143). Under the swap coin 22 of 24 bands are exactly flat and only the singlet pair moves, with $\sin w = \sin(\theta/2) g(K)$, g the mean of $\cos(K \cdot r)$ over the roots. So $\gamma = 0$ exactly (measured 7.6×10^{-12}), and $|v| \leq \sin(\theta/2) c/2$ at every K by Cauchy–Schwarz on the 2-design. Every excitation shares one limiting speed $c^* = c/2$, and nothing exceeds it at any of 4,096 momenta.

Built into the exact many-body rule, the one-hole matrix equals the float model to 4.9×10^{-15} . The sea stays one branch for 128 beats. Interaction frees nothing faster than c^* : two holes meeting release frozen weight only as free holes, and a fear, unmade at once into a store and two holes, walks one root every two beats. A hole's Born weight equals an empty-mesh love's at every slot, so gravity's sign holds (E-SPN-0145). The graviton's slide (E-GRV-0141) enters c only as a number mod p , and at the residue for $1/2$ its speed equals the slide's, so matter and gravity can share c^* .

What c^* costs: the stream's one root a beat becomes a microscopic bound that only 8% of a hole's weight reaches, not a physical speed. The husk light has not been rerun on this rule.

6. Binding

Motion under the candidate is a one-body statement. Matter is bound, and a bound composite must hold together, move isotropically under c^* , and have inertia equal to its energy. Under flat links separation costs nothing, and a level bound by contact alone sits inside the free two-hole continuum and decays (E-SPN-0145). The string must be supplied, and each attempt below narrows what it can be.

6.1. A cost on separation

The string closes only on charge 0 mod 3, so two holes cannot carry one. A hole on the love sea is exactly a love on the empty mesh, so the test is the love-fear meson on the empty mesh.

- **A phase string** (angle 0.0936 per unit of Steiner length, exact in $\mathbb{Z}[\omega][1/42]$) forms a composite at energy 1.5658 against 1.5705 predicted, isotropic to 9.5×10^{-9} , $R = 1.027$, moving at 0.17, 0.32 and $0.41c^*$ at $K = 0.4, 0.8$ and 1.2 . It is a resonance: at length $2m/\sigma$ the string has paid $2m$, and a member drops into 11 flat bands degenerate with its partner, where no Schwinger suppression applies [14] (E-SPN-0146).

- **A mass string**, the mixer angle growing with separation, closes that channel: the mixer acts only on a dock's uniform mode, the flat bands stay pinned at π , and every flipped channel tops out at $\pi - 2m_0$. The level holds exactly, fidelity $1 - 1.7 \times 10^{-11}$ over 256 beats with 1.8×10^{-18} reaching the edge, and moves isotropically. But it needs $m_0 > \arctan(1/3)$, reads the separation nonlocally, and gives $R = 1.568$ (E-SPN-0147).

Theorem 3 (The band sum) For the stream times any fixed dock matrix, the trace has no constant Fourier term because no root is zero, so the band phases sum to zero over the zone. At the swap point the moving pair averages to zero, so if the pair and the even flat bands are kept, the 11 odd flat bands stay at π . No dock-local unitary, in any ring, moves them alone (E-SPN-0148).

The motive was also wrong: the level carries only 0.16% per member in the flat bands, and R 's excess lives in the moving pair. On a line, where Gauss's law fixes the flux and a local string equals the instantaneous one, $R \rightarrow 1$ with the lattice mass $\tan(m)/m$ the only excess, falling as n^{-2} (measured slope -2.078), and binding moves R by at most 1.4×10^{-3} (E-SPN-0149). For an instantaneous string in d dimensions the inertia is $2m + (1 + 2/d)T$ against a rest energy of $2m + 3T$, so a cost-only string misses $(2 - 2/d)T$, which is the transverse inertia of a real string. The string needs dynamics of its own.

6.2. A gauge field on the links

- $\mathbb{Z}_3$ **flux**. A local flux register is exact, Gauss exact and vacuum inert, but it records each member's path, so a member tied to a charge walks the 24-regular tree, the universal cover of the D_4 root graph, and sits above the Kesten floor $\pi/2 - \arcsin(\rho \cos m_0)$, $\rho = 2\sqrt{23}/24$, at least 1.159 for every m_0 [5]. Measured edges match the floor to 10^{-3} (E-SPN-0150).

Theorem 4 (A) A magnetic plaquette piece is diagonal in link values and the hop is controlled by link values, so the two commute for every gauge group. The plaquette moves a member only through the electric piece (E-SPN-0151, E-SPN-0152).

Theorem 5 (B) In the flat sector a tied member walks D_4 exactly with its edge at m_0 , but there every link's flux is uniform, so for any finite group the string tension is exactly 0 (E-SPN-0151, E-SPN-0153).

Light and confined therefore needs a continuous confinement transition. $\mathbb{Z}_3$ in 4d has a first-order one [13], and in the $4 + 1$ bulk even $SU(2)$ does [12], so the string must live on the husk. The group the roots already form is $2T$: a fixed linear map sends the 24 roots onto the Hurwitz units, all 7 characters lie in $\mathbb{Z}[\omega]$, and a member carries the spinor 2. But $2T$ freezes while the string still has 0.82 of its strong-coupling energy per link, and $2O$ at 0.47 to 0.65 (E-SPN-0152), as finite subgroups of $SU(2)$ do [1, 4].

6.3. $2I$ on the husk

The husk lattice is cubic with 6 axis and 12 face-diagonal links per dock and 20 triangles, isotropic when a beat is $\sqrt{3/5}$ of an axis link. A deterministic one-loop crossing of $SU(2)$'s Gaussian free energy with the frozen branch puts $2I$'s hypercubic freezing at $\beta = 5.40$ against Monte Carlo's 5.70(20) [15]. On the husk $2I$ freezes at $\beta = 2.82$, matched hypercubic $\beta = 9.77$, where the string tension is 1.6×10^{-18} of its strong-coupling value (E-SPN-0153).

Result 1 A bounded 120-value register reaches $SU(2)$'s scaling region on the husk and loosens the string continuously. No continuous group is needed. The ring grows to $\mathbb{Z}[\omega, \varphi][1/210]$.

What fails is keeping. No ordered $2I$ register state is kept by the beat: the flat register keeps 0.813 per link per beat and the empty one 0.630 per face. A beat is expected to keep only locally maximally mixed states. What is proved is prethermal keeping [2]: at small beat angle δ a bounded register holds a state for time $\exp(c/\delta)$. Whether that holds the ordered $2I$ vacuum long enough is the one open question left in binding, and it is being run.

7. Open

- **Binding.** Whether the beat keeps the ordered $2I$ vacuum at small angle, then a light member bound by its string moving in 3d with $R \rightarrow 1$ (Section 6).
- **A temperature.** A forming horizon's spectrum is a power law, about $\omega^{-1.43}$, at every register size run, $C = 13$ to 49, so what shows is the growth's quench. Read late, a settled horizon makes 10^{-7} of the burst: it is a static wall with redshift capped at $\frac{1}{2} \ln C$. A bounded clock predicts a burst at formation and no steady evaporation unless $\ln C \gg 4\pi$ (E-GRV-0134, E-GRV-0136).
- **The bulk.** The warp, the radion's stabilization and brane bending are still taken from the continuum.
- **One light speed.** On the current coin a locked vibe tops out at $1/2$ a beat and every stable husk light at $1/\sqrt{6} = 0.408$, a theorem on each side (E-FRC-0250, E-SPN-0090). The candidate rule gives matter and gravity one c^* , and the light has not been rerun on it.
- **The electron in 3d, and g.** Three loves under the pass bind on one husk line (binding 1.079, spin-half share 0.982, E-SPN-0093), not yet across lines, and g needs the coupling to light.
- **The weak force.** The locked rule keeps P, C, T, CP and CPT under all 48 husk parities, and bulk chirality never reaches the husk, a theorem (E-FRC-0248, E-FRC-0249). It needs a spin apart from the slot and an orientation in depth. CP violation is open with it.
- **The value of α .** $\alpha = (s/24)\sqrt{3\kappa/2}$ per vibe, with s the drift's share of κ . No depth gives $1/137$, and nothing is read off (E-FRC-0242).
- **A bounded self.** Shared history gives one component of all 24,576 vibes or closed components of exactly 8 on one line, nothing between (E-SLF-0177, E-SLF-0178).
- **One rule for every row.** Many held results ran on an earlier rule and have not been rerun.

8. Changelog

Measured from *The Missing Amplitude* [8], which audited 834 experiments. The suite now holds 1,474.

8.1. The rule

- **2026-09-01 to 09-02.** The audit's follow-ups, the field content of the committed rule, and the turning weave (turningWeave), adopted 2026-09-02. Dispersion and coarse-bridge experiments.

- **2026-09-02 to 09-24.** The combined knit (E-FRC-0159), the fear beat as the source of amplitude (E-QTM-0118), and the light as column sums of bulk trits, compact $U(1)$ (E-FRC-0207, E-FRC-0244).
- **2026-09-25.** The strong-force branch: the gauge pieces and the string as a register.
- **2026-09-26.** Most pieces of Section 2 chosen: the $W(F_4)$ knit, the collision, the doublet lock, the meeting, the pass, the no-veto store and the coin, and the working vacuum with one C domain. Frozen domain walls were dropped as a cosmological liability.
- **2026-09-26 to 09-27.** The line law and its consequences: every mixer cascades, stores cannot join lines, the path key read the beat modulo 2^{16} and was replaced by a full-period key (E-MTH-0029).
- **2026-09-27.** Gravity through the depth and its horizons (Section 4). The moving-binding branch: a lone love's inertia goes to its energy as it gets light (1.006 at $n = 8$, E-SPN-0107), a string-bound pair's binding factor goes to 1 as the string loosens (1.018 at $D = 30$, E-SPN-0112), and along the joint limit a meson's inertia over energy reaches 1.003 (E-SPN-0113).
- **2026-09-27 to 09-28.** The open-pieces branch. The meson stays one particle at speed, with c within 0.14% of the walk's at $n = 16$, read through its Klein crossings, which an exact love-fear exchange symmetry sorts (E-SPN-0115). Spin 2 from the slide (E-GRV-0138 to E-GRV-0141), Randall-Sundrum (E-GRV-0137, E-GRV-0142, E-GRV-0143), gravity's sign on the sea (E-GRV-0144), the lineon no-go and its observational exclusion, the candidate rule for 3d motion, and the binding chain (E-SPN-0116 to E-SPN-0153).

8.2. Corrections

- **Results on earlier rules** stand as results about those rules. CP violation with CPT exact was measured on the turning weave, and the locked rule keeps CP exactly, so it is open again. The Cabibbo angle within four percent was never tested against a pre-registered null. No magnetic monopoles was an identity of an earlier light, and compact $U(1)$ admits heavy ones. The vacuum clock of period three belonged to the charge rule, and the working vacuum's cycle is 12 beats.
- **Gravity was once given as solved** from an area-law potential and an area-law invariant, both on a hand-written walk and graded stand-in. On the working vacuum the entanglement is a volume law at equilibrium (E-GRV-0083), and gravity read as the clock rate alone bends light by 1, not 2 (E-GRV-0088).
- **Within the open-pieces branch.** A reading of 31% frozen flat weight in the mass-string composite was the moving pair's own component, and the true figure is 0.16% per member (E-SPN-0148). A looser string's R of 1.751 was a different bound level (E-SPN-0111). "Finer layering shortens the range" was wrong: finer layering gives a gapless tower. A membrane horizon "fix" still left a frozen interior hidden, and was restated.
- **Instrument defects** caught before gate runs are disclosed in each experiment's header, and no gate was moved after a run.

A. Index of Experiments Cited

Identifiers as they appear in the suite’s catalog, with the verdict of the gated run. A fail is kept, and many of the fails below are the theorems the paper rests on.

identifier	verdict	what it measures
<i>The base and the rule</i>		
E-FND-0038	pass	three values and 4d are forced: the smallest signed alphabet, and the one root system with triality
E-SPN-0042	pass	the 24 roots carry the spinor double cover, a 2π turn gives -1
E-GMT-0025, 0027	pass	a spectral dimension near 3 on the cusp, and the honeycomb’s exact shells
E-RLT-0061	pass	a knit keeping all of $W(F_4)$ exists and keeps CPT
E-RLT-0097, 0099	pass	the doublet lock, and the meeting’s knot with CHSH $\sqrt{7}$
E-SPN-0090, 0091	pass	the coin theorem: its cost is exact lattice momentum
E-SPN-0092, 0093	pass	the contact theorem, and the electron bound under the pass on one line
E-RLT-0105	pass	the working vacuum on 51 of 51 cases
E-SPN-0098	pass	the line law: 6,144 of 6,144 line tones conserved
E-SPN-0094 to 0099	fail	every mixer the ring allows cascades through the vacuum
E-MTH-0029	pass	the full-period path key
<i>Light and the quantum</i>		
E-FRC-0207, 0244, 0245	pass	light as column sums of bulk trits, compact $U(1)$
E-FRC-0179, 0241, 0243	pass	two polarizations, Coulomb’s $1/(24\pi r)$, whole charges every beat
E-QTM-0118	pass	the fear beat is the one non-Clifford gate
E-QTM-0132, 0133, 0140	pass	Tsirelson in closed form and reached, Bell on exact values
E-QTM-0149 to 0154	pass	contextuality as fear, no-cloning, teleportation, GHZ, the pointer-basis theorem
E-FND-0146, 0147	pass	the second law and the arrow under the exact inverse
<i>Gravity</i>		
E-GRV-0079, 0080	pass	the radion: $1/r$ with no chosen sign, causal
E-GRV-0090	pass	depth found by summing one trit a link, Gauss exact on 43,008 checks
E-GRV-0092, 0093	pass	depth sets span as well as clock, light bends by 2
E-GRV-0104	pass	the pull arrives at $1.0023c$
E-GRV-0118, 0122, 0123	pass	a horizon at $2GM$, and Hawking’s redshift scaling
E-GRV-0131, 0135	pass	the membrane horizon’s area law and Bekenstein’s bound as dynamics
E-GRV-0132, 0134, 0136	fail	a forming horizon’s spectrum is a power law, and a settled one makes nothing
E-GRV-0137	fail	the scalar reaches $3/4$ of Randall–Sundrum, because the rest is spin 2
E-GRV-0138	partial	sliding unlabeled docks forces linearized Einstein–Hilbert
E-GRV-0139	partial	the spacetime slide: lapse and shift as multipliers

identifier	verdict	what it measures
E-GRV-0141	pass	the cubic slide fixes the graviton's speed at c and $\lambda = 1$
E-GRV-0142, 0143	pass	the tensor tower and the $4/3$ from a slide along the depth
E-GRV-0144	fail on one register	normal-ordered energy fixes gravity's sign on the sea
<i>Motion</i>		
E-SPN-0107, 0112, 0113	pass	falling alike: the walk factor and the binding factor each go to 1
E-SPN-0115	pass	the meson at speed, read through its Klein crossings
E-SPN-0116 to 0120	mixed	lineons: crossing pairs held, parallel lines independent, the star theorem, two hubs cascade
E-SPN-0121 to 0128	fail	six angles on 3d motion inside the rule, each closed with its cause
E-SPN-0126	fail	line-only motion is excluded by observation at every dock size
E-SPN-0130	partial	the Pauli-blocked mixer: the first free 3d motion near rest
E-SPN-0132, 0134	open, partial	link holonomy, and the candidate rule's audit
E-SPN-0136	partial	the frame mixer's cone is t , but its velocity stays in its frame
E-SPN-0137	open	a line-liquid vacuum: none exists locally
E-SPN-0140	fail, as derived	the 24-slot mixer: inert sea, isotropic mean band, velocity inside the 24-cell
E-SPN-0142	fail, as derived	the singlet is relativistic at exactly $c/2$
E-SPN-0143	fail on U4a	the swap coin: one c^* , $\gamma = 0$, the prime 7 for the mass
E-SPN-0145	fail on X4	the swap coin in the many-body rule: kinematics hold, nothing binds
<i>Binding</i>		
E-SPN-0135, 0139, 0141, 0144	fail, partial	three holes on a slab under the old coin: a dense family of bent levels
E-SPN-0146	partial	a phase string: a composite that moves, and leaks through the flat bands
E-SPN-0147	fail on R	a mass string: an exact bound composite that moves in 3d, with $R = 1.568$
E-SPN-0148	fail, as derived	the band-sum theorem: no dock piece moves the flat bands
E-SPN-0149	pass	on a line $R \rightarrow 1$, the lattice mass the only excess
E-SPN-0150	fail, as derived	a $\mathbb{Z}_3$ link flux: members walk the tree, above the Kesten floor
E-SPN-0151	fail, as derived	a $\mathbb{Z}_3$ plaquette: theorems A and B
E-SPN-0152	fail, as derived	$2T$ from the roots freezes with a heavy string, and the bulk cannot confine smoothly
E-SPN-0153	fail on H3	$2I$ on the husk reaches the scaling window, and no ordered state is kept

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